MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. Rate of x^2 between 1 and 2:
   - A. 2
   - B. 3
   - C. 1
   - D. 4
2. If $f(1) = 4$, $f\prime(1) = 2$, the tangent is:
   - A. $y = 2x + 2$
   - B. $y = 4x + 2$
   - C. $y = 2x + 4$
   - D. $y = x + 3$
3. The derivative at a point is
   - A. a function
   - B. an area
   - C. an x-coordinate
   - D. a real number if the derivative exists
4. For x^2 at 3, the derivative at that point is:
   - A. 9
   - B. 3
   - C. 6
   - D. 12
5. The approximation of $f(a + h)$ is
   - A. $f(a)f\prime(a)$
   - B. $f(a) + hf\prime(a)$
   - C. $f(a) + f\prime(h)$
   - D. $hf(a)$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $\sqrt{x}$ at 0 is
   - A. differentiable with derivative 0
   - B. differentiable with derivative 1
   - C. not differentiable
   - D. not defined
2. $\dfrac{|h|}{h}$ for $h < 0$ equals
   - A. -1
   - B. 1

C. 0
D. h

3. A tangent of slope 0 is
   A. vertical
   B. oblique
   C. impossible
   D. horizontal
4. Position in m and time in s: the unit of the derivative is
   A. m^2
   B. $\mathrm{m\,s}^{-1}$
   C. $\mathrm{s\,m}^{-1}$
   D. m
5. For f differentiable at a, replacing $f(a+h)$ by $f(a)+hf\prime(a)$ is in general
   A. an exact identity
   B. a definition of h
   C. a local approximation
   D. always false, even for an affine function

True or false? Justify.

1. A tangent can never cross a curve again.
2. The absolute value function is differentiable at 0.
3. An affine function $mx+p$ has derivative m at every real number.

Key point · **Self-assessment.** I give the domain and h nonzero; I read a slope; I use the point of tangency; I distinguish exact from approximate.

For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.